Privacy-Preserving and Explainable Heart Disease Risk Prediction Using Federated Learning and Explainable AI

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Heart disease remains the leading cause of death worldwide, making early and reliable risk prediction crucial for timely clinical intervention. Traditional machine learning methods encounter two unresolved challenges: they often require centralizing sensitive patient data, which raises privacy and regulatory concerns, and they depend on black-box models whose predictions are not interpretable by clinicians. This paper introduces a unified system that addresses both issues by integrating federated learning with explainable artificial intelligence. Five supervised models are assessed individually and in combination using soft voting and stacking, with a stacking ensemble of logistic regression, random forest, and XGBoost chosen as the production model. Shapley Additive explanations are used to determine global feature importance, while local interpretable model-agnostic explanations offer patient-level insights that clinicians can understand. Using the UCI Cleveland dataset, which includes 303 patients and 13 clinical features, the stacking ensemble achieved about 91% accuracy, and the federated hybrid model matched this performance without sharing raw patient data.

Keywords—*federated learning, explainable AI, SHAP, LIME, heart disease prediction, stacking ensemble, privacy-preserving machine learning, clinical decision support*

I. INTRODUCTION

Cardiovascular disease is one of the leading causes of death worldwide [1]. Identifying people who are at risk at an early stage can help doctors provide timely treatment and reduce the chances of serious complications. Machine learning (ML) offers a useful way to analyse clinical data and identify patterns that may be difficult to recognise through traditional methods [2]. However, two major challenges need to be addressed before these techniques can be widely used in healthcare.

The first challenge is patient data privacy. Many machine learning systems require medical records from multiple hospitals to be collected on a central server for model training. This can expose sensitive patient information to security risks and raise ethical and regulatory concerns. Federated learning provides an alternative approach by allowing each hospital to train its model locally and share only model updates with a central server, without transferring raw patient records.

The second challenge is explainability. Deep learning models and ensemble methods can produce accurate predictions, but understanding how they reach their decisions is often difficult. This lack of transparency may reduce clinicians' confidence in using these systems. Explainable AI (XAI) techniques, such as SHAP and LIME, help address this issue by showing which clinical features influence a prediction, both across the overall dataset and for an individual patient.

This research combines federated learning and explainable AI in a single framework for heart disease prediction, with a focus on accuracy, patient privacy, and interpretability. The main contributions include a stacking ensemble that achieves approximately 91% accuracy on the UCI Cleveland dataset, a federated hybrid architecture designed to achieve comparable predictive performance without sharing raw patient data, and an explanation layer that uses both SHAP and LIME to help clinicians understand model predictions at the population and individual patient levels.

II. RELATED WORK

There are other work on privacy/explainability for heart disease prediction. Few combine them. Sathi et al. [7] adopt explainable federated learning for multi-class diagnosis of heart disease based on ECG fiducial features while protecting hospitals' privacy, incurring high communication cost. Rezk et al. [8] use ensemble-based XAI techniques to pinpoint heart attack risk factors, but data are assumed to be centralized. Some interpretable ML models, such as SVM, logistic regression, and decision tree that identify important features, can yield about 88% accuracy without any privacy protections. A tutorial-style description of a straightforward combination of ML, XAI, and FL models to predict heart attack risk is provided by Agrawal [9]. See TABLE I.

TABLE I. SUMMARY OF RELATED WORK IN FEDERATED AND EXPLAINABLE HEART DISEASE PREDICTION

Study	Method	Results	Gap
Sathi et al. (2025) [7]	FL + XAI across hospitals	About 92% acc.; privacy preserved	High comm. overhead

Study	Method	Results	Gap
Interpretable ML (2023)	SVM, LR, DT with feature importance	About 88% acc.; explainable	No privacy
Rezk et al. (2024) [8]	Ensemble + XAI	Improved acc.; risk factors found	Centralized data only

Throughout, systems that provide explanation leverage centralized data while systems that protect privacy via federation seldom do and the handful that do both reveal that they impose significant communication requirements. To our knowledge, no system currently exists to provide privacy and explanation in a single light-weight pipeline and that is the focus of our work.

III. Problem Statement

Two categories of risk currently hinder the use of ML for predicting heart disease in clinical settings.

- Black-box models. Ensembles and deep nets deliver predictions but do not explain what they are based on yet, this is regarded as permissible in the clinical setting, where there must be trust.
- Centralized data risk. Traditional pipelines aggregate patient data from several hospitals on one server, creating serious risk of data breaches, unauthorized access.

Because of this, a system for authentic multi-hospital collaboration in knowledge discovery would have to offer:

(i) reliable predictions,

(ii) autonomous, privacy-preserving training,

(iii) clear and intelligible interpretation, and

(iv) compliance with ethical and legal constraints.

IV. Proposed System

The proposed system is built on three interconnected pillars: a machine learning module that produces an accurate risk prediction, a federated learning module that trains the model without centralizing patient data, and an explainable AI module that shows clinicians why each prediction was made. Together they form a single pipeline that is accurate, privacy-preserving, and transparent.

A. Machine Learning Module

The first pillar trains and assesses five supervised classifiers using the preprocessed clinical features. Logistic regression (LR) estimates disease probability as a sigmoid function of a weighted sum of the features, providing an interpretable baseline. The support vector machine (SVM) identifies the maximum-margin boundary between patients with and without disease, and uses a kernel function to separate classes that are not linearly separable. K-nearest neighbours (KNN) assigns a patient to the majority class among the most similar patients in the training set. Random forest (RF) aggregates multiple decision trees, each built from a random subset of patients and features, which helps reduce the overfitting seen in individual trees. XGBoost (XGB) [10] constructs trees sequentially, with each new tree addressing the errors of previous ones under a regularized objective.

The individual models each make different types of errors, which is why we ensemble the models in two different ways. Soft voting takes an average of the predicted probabilities of LR, RF, and XGBoost. Stacking [11] takes the method even further: the predictions of the base models are the features for a second level LR meta-learner, which assigns the new models a final weight. Hyperparameters are optimized with RandomizedSearchCV, GridSearchCV, and Optuna [12]. The stacking ensemble resulted in the highest accuracy, which we use for our production model.

B. Federated Learning Module

The second pillar eliminates the pooling of patient records. Every hospital that participates operates as a client and keeps its data on-site, training a local deep neural network (DNN) and an XGBoost model.

From the server, the current global model is shared to each hospital.

Each hospital will train only its own patients.

Only model weights are returned by each hospital, not raw records.

The server computes the new global DNN by taking an average of the weights.

With this feedback loop, the global model learns from all institutions, but each patient record stays in its hospital. Since only weights are transmitted, the risk of breaches, illicit access, and noncompliance with regulations is much lower.

C. Explainable AI Module

To that base, the third pillar appends a clinical lens: one of two methods to explain any individual prediction. SHAP [5] calculates a Shapley value for each feature: the average amount the feature moved the prediction in the direction of high or low risk, across all feature subsets; then it sums exactly to the difference between the patient's prediction and the average prediction. Summary across patients yields a global feature importance ranking (summary plot), looking at the patient population altogether. Summarising individual predictions (waterfall plot) shows us how each feature value changed the risk prediction in that patient. LIME [6] is more local: taking a single patient, it perturbs their feature values, measures the change in the prediction, and fits a simple (local) weighted linear model to that patient to tell us, which features made that patient more or less risky.

V. Methodology

The method is in steps.

A. Data Preprocessing

Preprocessing includes data cleaning, handling of missing value with 3 bounds and IQR normalization feature scaling with StandardScaler and traintest split.

B. Supervised ML Training

With the training set, I train and test out logistic regression, support vector machine, K nearest neighbours, random forest and XGBoost. (50+ trials)

C. Ensemble and Final Model Selection

Comparison of soft voting (LR+RF+XGB) and a stacking ensemble using an LR meta learner.

D. Federated Learning

The server combines the model weights to form a global DNN and sends the new parameters to start the next epoch.

E. Explainability

SHAP has both global feature importance and waterfall plots LIME offers patient level explanation and feature contribution.

VI. System Architecture and Workflow

The architecture is based on a client–server federated learning approach, As shown in Fig. 1, the proposed system uses a client–server architecture based on federated learning. Each participating hospital works with its own data and trains DNN and XGBoost models locally, so patient records do not need to be shared with other hospitals or the central server. Instead, the hospitals send only their model updates to the central server, where the updates are combined to improve the global model. The updated model is then sent back to the hospitals for the next round of training. This process continues over multiple rounds, helping the model learn from different hospitals while keeping patient data within their respective institutions. The system uses the UCI Cleveland Heart Disease dataset, which contains 303 patient records with 13 clinical features, including age, blood pressure, cholesterol levels, ECG results, and maximum heart rate

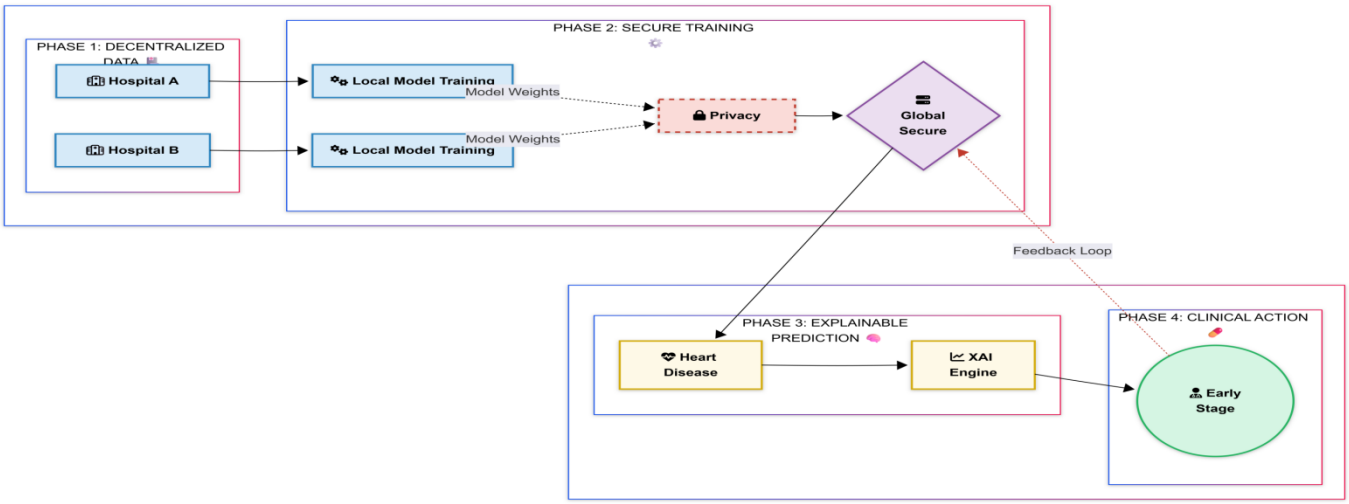


Fig. 1. System architecture and workflow: decentralized training at each hospital, aggregation of model weights into a global model, explainable prediction, and a feedback loop to clinical action.

VII. Results and Discussion

A. *Predictive Performance*

Table II compares the accuracy of each individual model and of the ensemble strategies.

TABLE II. Model Accuracy on the UCI Cleveland Dataset

Model	**Accuracy (%)**
Logistic Regression	80.0
SVM	82.0
KNN	78.0
Random Forest	87.0
XGBoost	88.0
Voting Ensemble	89.0
Stacking Ensemble	*91.0*

The stacking ensemble, which combines Logistic Regression (LR), Random Forest (RF), and XGBoost, achieved the highest accuracy of approximately 91%. It performed better than the individual models and the voting ensemble. The federated hybrid model also achieved a similar level of accuracy without requiring hospitals to share raw patient records. These results suggest that the proposed approach can maintain predictive performance while protecting patient data.

B. *Explainability with SHAP and LIME*

SHAP analysis also revealed several relevant predictive factors that included chest pain type (0.92), maximum HR (0.78), ST depression (0.71), age (0.64), cholesterol (0.52), and BP (0.45) (Fig. 2). Whereas SHAP focuses on overall feature contribution, LIME explains individual cases, where individual clinical features and their positive or negative contribution to an individual prediction are displayed (Fig. 3). Also, the feature heatmaps help to interpret whether features steer predictions to a higher or lower risk, which makes interpretation easier for the doctors and enables comparison with their own opinion. For example, an individual prediction where a diagnosis of high risk of heart disease is presented is

mainly dictated by ST depression (+0.31), chest pain type (+0.28) and age (+0.19), making the model's prediction easily understandable. This simple interpretation hopefully contributes to clinicians' trust in the model.

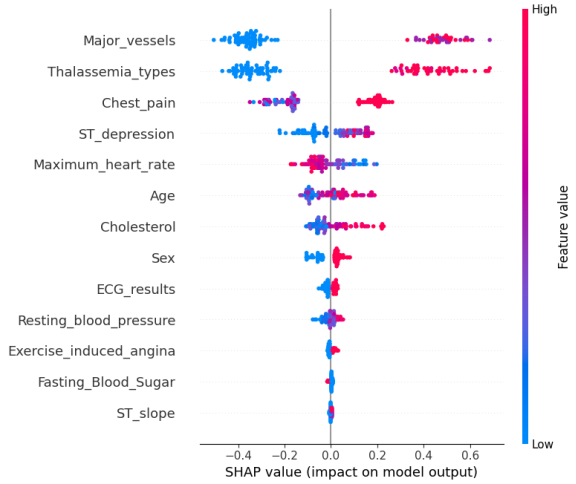


Fig. 2. SHAP summary plot of global feature importance.

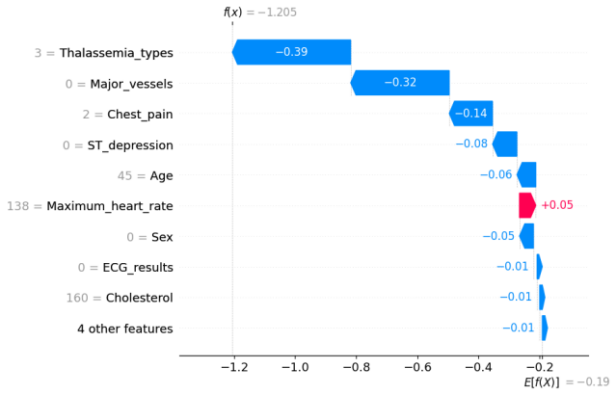


Fig. 3. Patient-level explanation: per-feature contributions to a single prediction (waterfall plot).

C. Discussion

The team found that the stacking ensemble outperformed the models individually, but the federated hybrid model had the same level of accuracy as training centrally, without sharing raw patient data. The team can also use SHAP and LIME to explain feature importance not just across the dataset but on each individual patient record. The work is restricted by a single open dataset, with 303 individual patients' data. To verify the real-world applicability of the proposed system, the researchers will test the models on data collected from several hospitals.

VIII. Conclusion

This study developed a heart disease prediction framework that combines federated learning with SHAP and LIME to make the model's predictions easier to understand. Hospitals can train models using their own patient data without sharing raw medical records, while the federated hybrid model achieves accuracy comparable to centralized training. The stacking ensemble, which combines Logistic Regression (LR), Random Forest (RF), and XGBoost, achieved approximately 91% accuracy on the UCI Cleveland dataset. SHAP helps identify the clinical features that influence predictions across the dataset, while LIME explains the factors behind an individual patient's result. Together, these methods can help doctors better understand the model's decisions. Overall, the proposed framework demonstrates a practical approach to heart disease prediction that considers accuracy, patient privacy, and transparency, although further validation using data from multiple hospitals is needed before real-world clinical use.

IX. Future Work

- *Real-time patient data.* Integrate live monitoring streams from wearables and IoT sensors for continuous, dynamic risk scoring.
- *Advanced deep learning.* Extend the framework with hybrid CNN-LSTM or transformer-based models to capture temporal patterns in ECG and multi-modal data.
- *Multi-disease framework.* Scale the federated system to diabetes, stroke, and hypertension as a comprehensive preventive AI platform.
- *Real hospital deployment.* Pilot the system in hospital networks to evaluate scalability, communication efficiency, and real-world FL performance.

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